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1 – DESCRIPTION

Troubleshooting EPI ghosting problems is often very challenging, given that EPI has so many mechanisms that contribute to ghosting. This checklist is designed to provide a structured approach to correcting EPI ghosting problems in EXCITE (10.0) systems (do not use this document to troubleshoot LX (8.X, 9.X) systems). Here is how to check what revision software (compare the grayed number) your system has loaded:

```
{sdc@t15}[1] whatRev
Hostname : t15
Build number for MrpApps is 10.0_M4_0309.a
Build number for PostSdC is 10.0_M4_0309.a
Build number for cclass is 10.0_M4_0309.a
Build number for devenv is 10.0_M4_0309.a
Build number for driverSupport is 10.0_M4_0309.a
Build number for insite is 10.0_M4_0309.a
Build number for install is 10.0_M4_0309.a
Build number for mrtest is 10.0_M4_0309.a
Build number for os_cd is 6_15.4.a
MR Software release: 10.0_0309a
{sdc@t15}[2]
```

Note
Revision may be later than what is shown.

2 – HOW TO USE THIS DOCUMENT

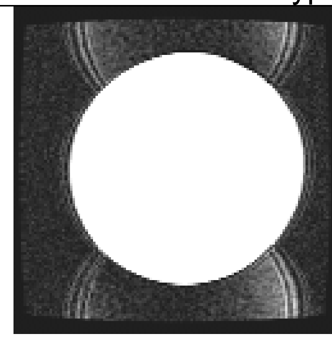
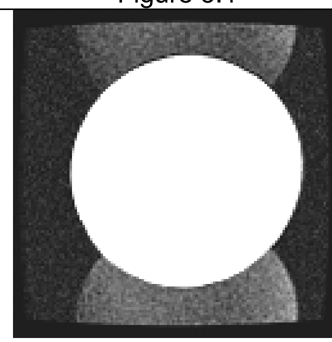
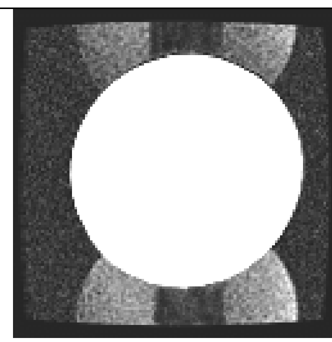
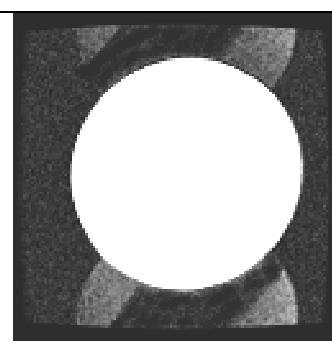
Start by comparing the customer's ghosted images to the examples in Section 3. Each example has a list of things to check to eliminate that particular type of ghosting, with the most likely causes (or checks taking the least amount of time) listed first. Go to Section 4 for instructions on how to perform each check. If you are instructed to run EPT then run it as described at the top of Section 3.

3 – TROUBLESHOOTING USING EPT

EPT is typically run without the TLT Head Loader (that is the only way the standard foam holder for the 10 cm NiCl sphere will fit in the head coil). To get a better assessment of clinical EPI image quality, use patient comfort pads, towels etc. to center the 10 cm NiCl sphere (46-317586G1) in the TLT Head Loader (46-287899G1), put the sphere and the loader in the head coil and run the Head B-zero Image test as described in the EPT Service Procedure.

3-1 Different Types of EPI Ghosts

The following section contains examples of different types of ghosting typically seen in EPI images. Compare the images from the customer to the images below and execute the checks (Section 4) listed for the ghosting type most closely resembling your case. If the customer's images contain several types of ghosting, start with the most dominant type.

3.1 "Edge" ghosting – normally seen with ramp-sampled EPI prescriptions Check (Section 4): 6, 8, 10	 Figure 3.1
3.2 B0, "Constant" or "Flat Field" EPI ghosting – is caused by a constant phase error between even & odd EPI echoes. The ghost intensity is approximately uniform Check (Section 4): 1, 2, 3, 4-2, 5, 9, 10, 11	 Figure 3.2
3.3 "Linear" EPI ghosting – similar to a B0 ghosting in appearance with the addition of a vertical signal void. Linear EPI ghosting is caused by a residual delay between even & odd echoes. Increased linear ghosting normally indicates a gradient problem. Check (Section 4): 3, 4-1, 4-2, 6, 7, 8, 9	 Figure 3.3
3.4 "Tilted Linear" EPI Ghosting - linear EPI ghosting with a visible tilt in the signal void that indicates an uncompensated cross-term eddy current generated by the readout gradient that maps to the phase encode gradient. Check (Section 4): 3, 8	 Figure 3.4

4 – EPI GHOSTING CHECKS

4-1 White Pixels

What to do: Run EPI White Pixel in the Head Coil and make sure you have no white pixels with the TNS disabled.
Why? White pixels during EPI Reference scan may cause failures in the linear and constant compensation, leading to artifacts in the final image.

4-2 Shorted RF Shield

What to do: Use Report Manager to look at your EPT B-zero Dither results. If, for one or more axes, no part of the plot is visible in the plot window (the entire graph is outside the plot window), call the OnLine Center for support.
Why? A very large phase dither (>40 degrees may be bad, >60 degrees is almost certainly bad) needed for all echo spacings for one or more axes may be a sign of a short circuit in the RF shield.

4-3 Swap Phase/Frequency

What to do: Select the protocol that the customer is complaining about, change the Phase/Frequency selection, and scan. If the ghosting is less, you have a problem with the gradient axis that was originally used for phase encoding and you should check Grafidy and stability for that axis.

4-4 GP Firmware

The Gradient Processor (GP) Module is located in the top of the ACGD Cabinet directly below the fan. There are two varieties of GP Module. The earlier GP Module is marked with GE part number 2248992. The later GP2 Module is marked with GE part number 2248992-3. GP and GP2 Modules that have not already been upgraded by FMI 60637 (tentatively scheduled for release FW 10) need to be checked in order to confirm that the correct firmware is installed. GP or GP2 Modules in need of the new EEPROM or capacitor/jumper firmware will receive this hardware as follows:

- Systems upgrading from 4.X, 5.X, or LX to Signa EXCITE that are also receiving a new ACGD Cabinet and, at the same time, receiving 10.0 M4 software will receive the new GP2 Module with the firmware already installed at the factory.

- New Signa EXCITE installations receiving 10.0 M4 software will receive the new GP2 Module with the firmware already installed at the factory.

- Systems with an ACGD Cabinet upgrading from LX to Signa EXCITE and at the same time receiving 10.0 M4 software will receive the firmware upgrade kit and instructions in the Signa EXCITE Upgrade Collector kit. The GP or GP2 Module already on site inside the ACGD Cabinet will be upgraded by the Field Engineer.

- All other Signa EXCITE systems will receive the firmware upgrade kit and instructions as part of FMI 60637 (10.0 M4 upgrade). The GP or GP2 Module that is already on site and inside the ACGD Cabinet will be upgraded by the Field Engineer as part of the FMI.

4-4-1 GP and GP2 Module EEPROM Check

These GP and GP2 Modules will need the new U108 EEPROM:

- All GP’s (2248992) without deviation M-179-02 applied to front of unit

- GP2’s (2248992-3) with Revision 0 applied to the front of the unit

4-4-2 GP and GP2 Module Capacitor/Jumper Check

These GP and GP2 Modules will need the new Capacitor/Jumper Assembly:

- All GP’s (2248992)

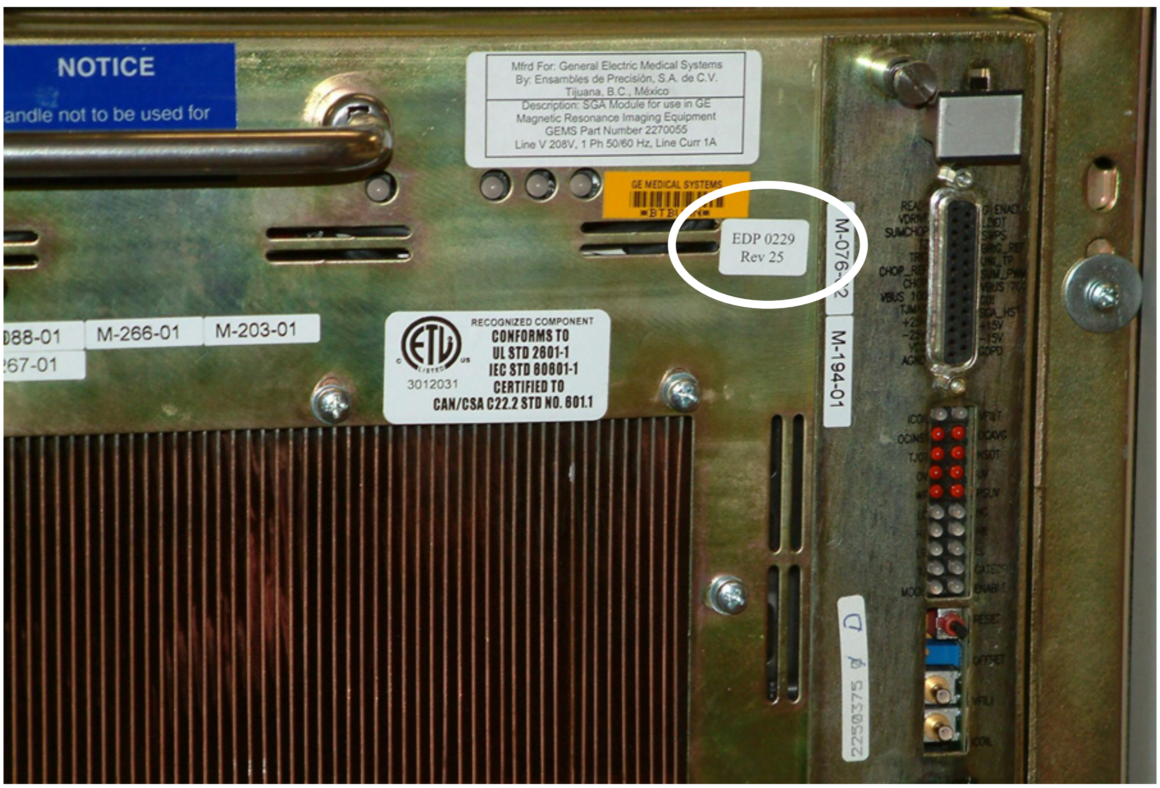
- All GP2’s (2248992-3) with Revision 0, 1, or 2 applied to the front of the unit

4-5 FETs On SGA Driver Boards

Scalable Gradient Amplifiers (SGA's) from Teledyne with date codes from April to August 2002 have a problem with the MOSFET on the gate driver board. Teledyne SGA's are recognizable by their gold color and a small sticker with "EDP 02XX Rev 25" where the XX is the fiscal week the unit was made. The problem date codes are between **0212** and **0239**.

NOTE

IF your SGA Date code falls within the affected range then check for Manufacturing Deviation Stickers M-348-02 or M-356-02 applied to the front of the unit. If it has either of these 2 deviations then the Gate Driver Board has been already upgraded and is **NOT** at risk. IF your SGA Date code does fall within the affected range and does not have the correct manufacturing deviation stickers then contact your FMI Coordinator and obtain FMI 60641. You will need to order one FMI 60641 kit for each board that needs to be replaced. FMI 60641 is tentatively scheduled for release FW10. These boards will not be available before FMI 60641 is released.



FRONT VIEW OF SGA
ILLUSTRATION 4-1

4-6 L-coil values

Issue: Some sites may have the wrong L-coil values. Run Healthpage and see if it is reporting any failures concerning the L-coil values.

Fix: Perform the L-coil Calibration procedure documented in the Grafidy3 Calibration process and recheck eddy current calibrations using Grafidy3. It may only be necessary to just check and improve the individual terms that need adjustment.

4-7 Aydin Monitor

One site has reported ghosting caused by the presence of an Aydin Monitor in the scan room. If your site has such a monitor, try moving it out of the scan room and see if the ghosting is reduced.

4-8 Grafidy

Verify that your Eddy Current compensation is done to the best possible level of compensation, both for long and very long time constants, on-axis and cross-terms. Short time constant compensation should not be run, as it is not expected to yield better results than the system defaults.

4-9 Coherent Noise

Run SPT Coherent Noise Test and verify that no coherent noise is present on your system.

4-10 HSS

HSS is used to make sure that environmental vibration or electrical disturbances (e.g. moving metal) are not the source of the ghosting problem.

4-11 Bandpass Asymmetry Correction Characterization

The goal of Bandpass Asymmetry Correction Characterization (BACC) is to remove phase and magnitude asymmetries in EPI data that have been introduced by the analog filters of the receiver. The current BACC technique, however, is also sensitive to the phase and frequency response of the exciter. This can lead to erroneous BACC vectors and higher levels of EPI ghosting.

A simple test can be performed to identify if this is a problem by acquiring one EPI scan with the BACC enabled and one with BACC disabled. If the ghosting is reduced when BACC is disabled then the BACC vectors are corrupted.

Note

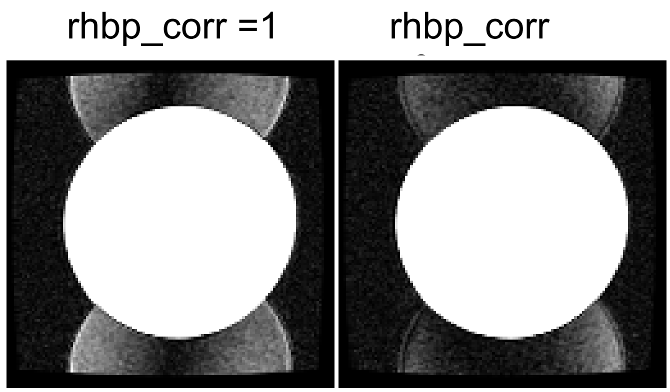
Use Patient ID “geservice” to put the scanner in research mode. This is required to enable you to modify CVs.

Prescription: Axial, SE-EPI, 1 shot, minFull TE, 1000 ms TR, (opuser0 = 1) Ramp Sampling = ON, 24 cm FOV, 5 mm thick slice, 0 gap, 64x64 matrix, 1 NEX, 1.0 phaseFOV, 1 slice at isocenter.

Scan 1: Download the above prescription and press Scan

Scan 2: Under “Research Options”, select “Display CVs’” and set the CV rhbp_corr = 0 to disable the bandpass correction. Re-run Reference Scan. and press Scan

Compare the ghost level between the 2 scans. Below is an example of EPI ghosting that was increased due to corrupted BACC vectors.

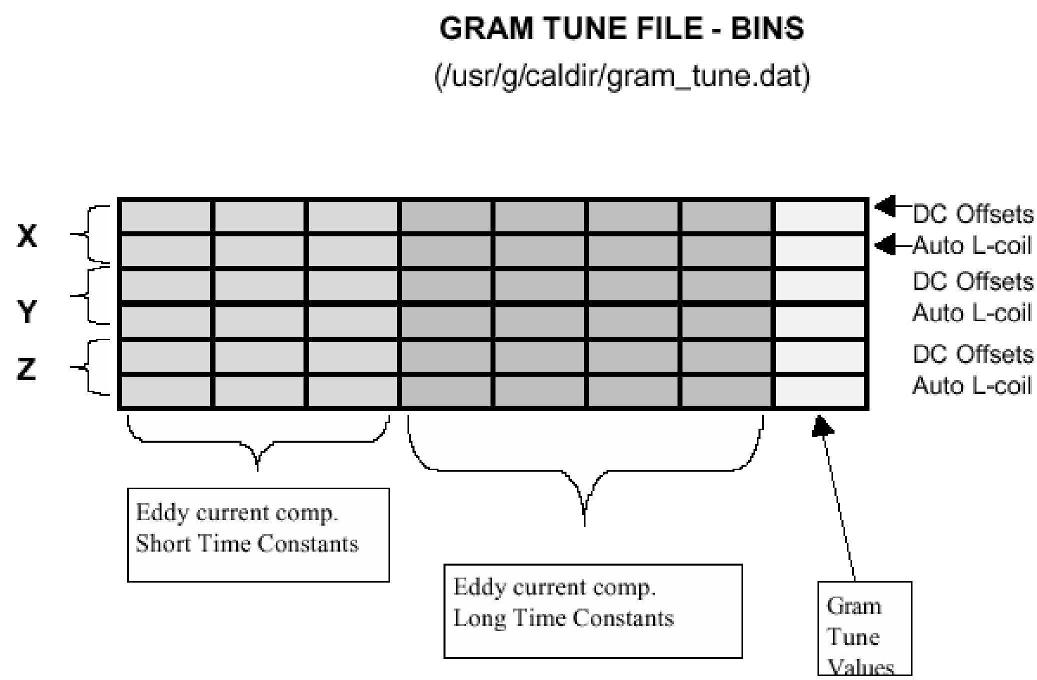


EPI GHOSTING COMPARISON
ILLUSTRATION 4-2

If you believe that the BACC files are corrupted then run the BACC procedure again and repeat the two scans. Repeat the comparison process described above.

APPENDIX

Eddy Current Compensation File Formats



GRAM_TUNE.DAT FILE EXAMPLE
ILLUSTRATION A

EDDY CURRENT CALIBRATION FILE BINS

(grafidyx.cal, grafidyy.cal and grafidyz.cal)

taus	*alphas*		*grafidyx.cal*	*grafidyy.cal*	*grafidyz.cal*
0.000000	0.000000	B0 Long Time Constants	X	Y	Z
0.000000	0.000000		X	Y	Z
0.000000	0.000000		X	Y	Z
0.000000	0.000000		X	Y	Z
0.000000	0.000000	On-Axis Linear and Cross-Term Long Time Constants (4 time constants per term.)	**X - X**	Y - X	Z - X
0.000000	0.000000		**X - X**	Y - X	Z - X
0.000000	0.000000		**X - X**	Y - X	Z - X
0.000000	0.000000		**X - X**	Y - X	Z - X
0.000000	0.000000		X - Y	**Y - Y**	Z - Y
0.000000	0.000000		X - Y	**Y - Y**	Z - Y
0.000000	0.000000		X - Y	**Y - Y**	Z - Y
0.000000	0.000000		X - Y	**Y - Y**	Z - Y
0.000000	0.000000		X - Z	Y - Z	**Z - Z**
0.000000	0.000000		X - Z	Y - Z	**Z - Z**
0.000000	0.000000		X - Z	Y - Z	**Z - Z**
0.000000	0.000000		X - Z	Y - Z	**Z - Z**
0.000000	0.000000	On-Axis Short Time Constants (4 time constants.)	X	Y	Z
0.000000	0.000000		X	Y	Z
0.000000	0.000000		X	Y	Z
0.000000	0.000000		X	Y	Z
0.000000	0.000000	Very Long Time Constants 2 time constants per term.	**X - X**	Y - X	Z - X
0.000000	0.000000		**X - X**	Y - X	Z - X
0.000000	0.000000		X - Y	**Y - Y**	Z - Y
0.000000	0.000000		X - Y	**Y - Y**	Z - Y
0.000000	0.000000		X - Z	Y - Z	**Z - Z**
0.000000	0.000000		X - Z	Y - Z	**Z - Z**
0.000000	0.000000	Very Long B0 Time Constants	X	Y	Z
0.000000	0.000000		X	Y	Z

GRAFIDY CAL FILE EXAMPLES

ILLUSTRATION B

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	FEB 6, 2003	Don Thome	Initial Release
1	FEB 25, 2003	Don Thome	Made corrections to Note in Section 4-5.
2	FEB 28, 2003	Don Thome	Corrected errors concerning M3 and M4 software.